



Scrum Review for Sprint 6

StockMaster3000

Key Implementations

1



Issues identified
during static code
analysis.

**SpotBugs
Findings**

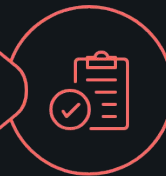
2



Improvements made
for better code
quality.

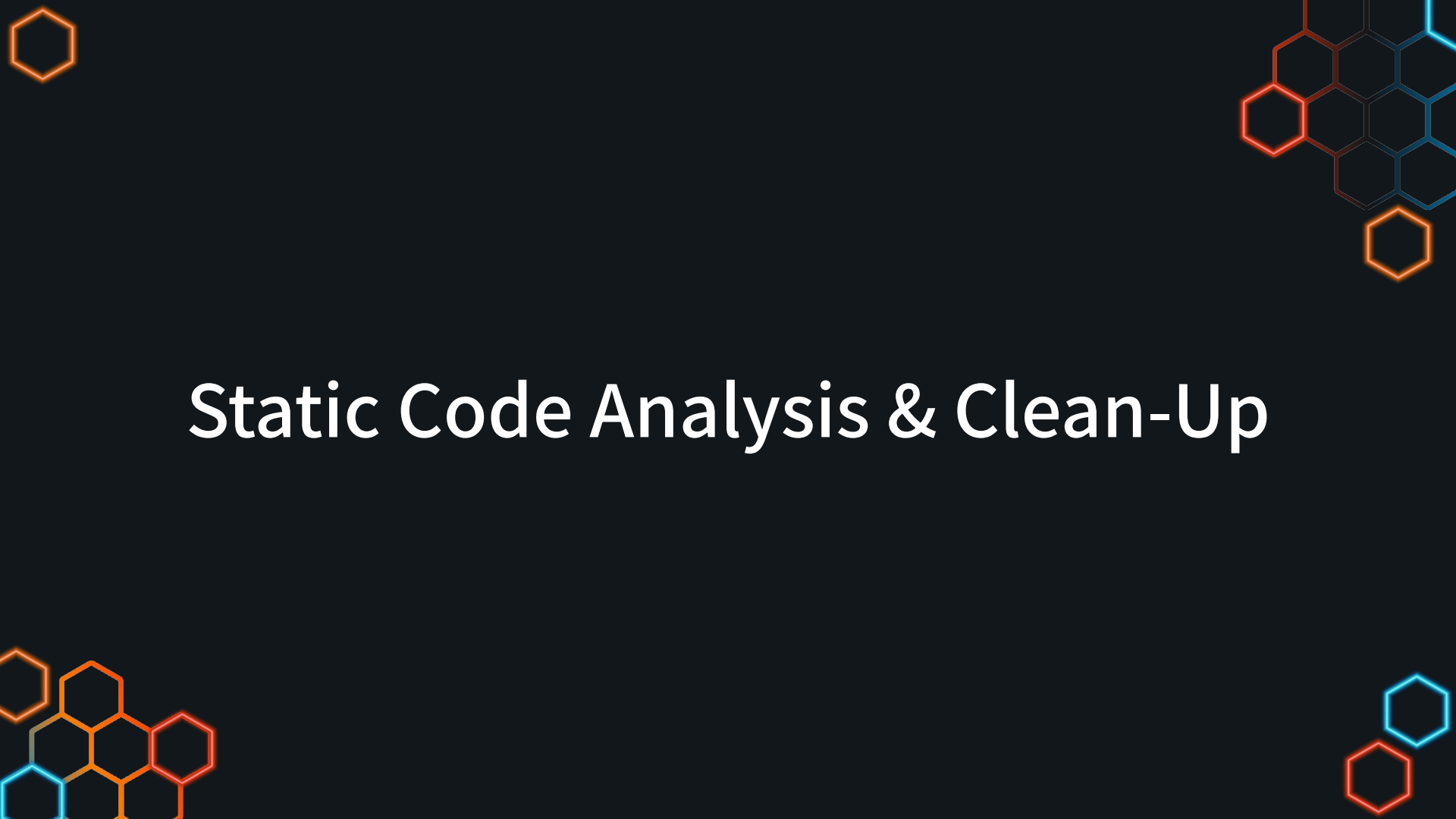
**Refactoring
Actions**

3



Ensuring readiness
for acceptance
testing.

**Acceptance
Test
Preparation**

The slide features a dark blue background with decorative hexagonal patterns in the corners. The top-left corner has a single orange hexagon. The top-right corner has a cluster of orange and blue hexagons. The bottom-left corner has a cluster of orange and blue hexagons. The bottom-right corner has a small cluster of orange and blue hexagons.

Static Code Analysis & Clean-Up

Rules: Google Checks

Checkstyle found 365 item(s) in 38 file(s)

BarChart.java : 4 item(s)

- Extra separation in import group before 'java.util.Arrays' (9:1) [CustomImportOrder] <Google Checks>
- Missing a Javadoc comment. (38:3) [MissingJavadocMethod] <Google Checks>
- Missing a Javadoc comment. (133:3) [MissingJavadocMethod] <Google Checks>
- Line is longer than 100 characters (found 122). (134:0) [LineLength] <Google Checks>

Category.java : 7 item(s)

- Using the '.*' form of import should be avoided - jakarta.persistence.*. (3:27) [AvoidStarImport] <Google Checks>
- Extra separation in import group before 'java.util.ArrayList' (5:1) [CustomImportOrder] <Google Checks>
- Missing a Javadoc comment. (9:1) [MissingJavadocType] <Google Checks>
- Missing a Javadoc comment. (23:3) [MissingJavadocMethod] <Google Checks>

- (12:1) [MissingJavadocType] <Google Checks>
- (15:3) [MissingJavadocMethod] <Google Checks>
- (17:3) [MissingJavadocMethod] <Google Checks>
- (19:3) [MissingJavadocMethod] <Google Checks>
- (21:3) [MissingJavadocMethod] <Google Checks>
- (23:3) [MissingJavadocMethod] <Google Checks>
- (26:3) [MissingJavadocMethod] <Google Checks>

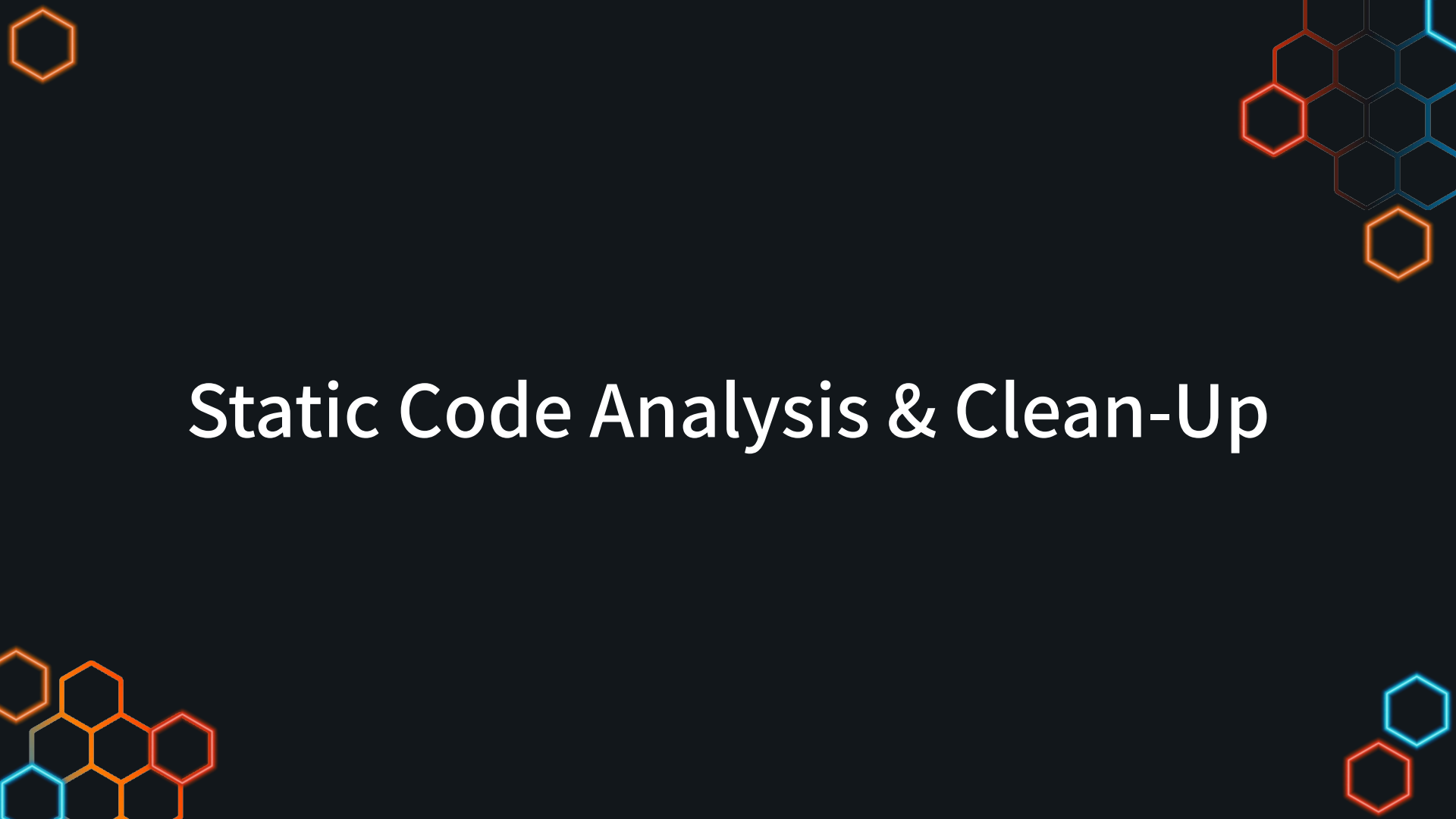
Rules: Google Checks

Checkstyle found 12 item(s) in 4 file(s)

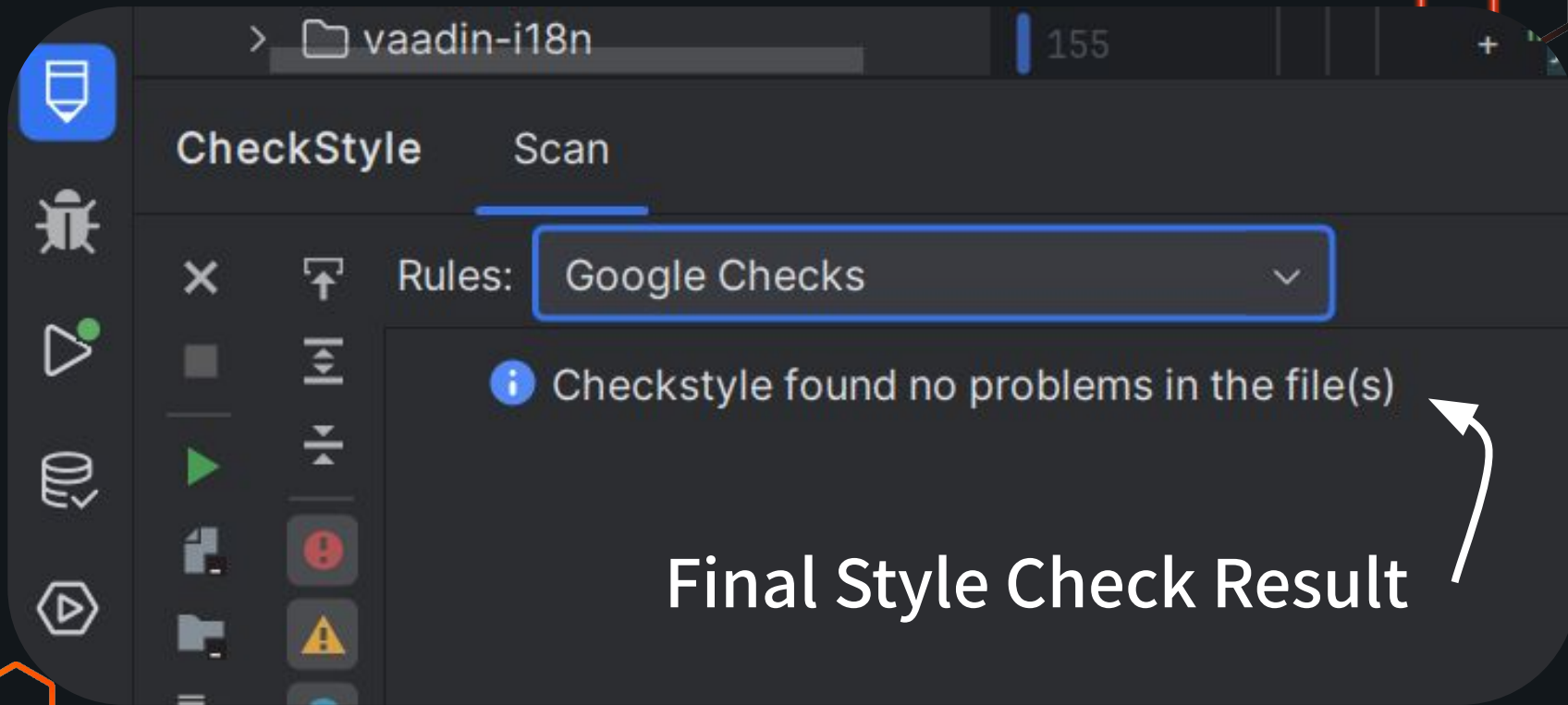
LineChart.java : 4 item(s)

- Abbreviation in name 'YAxel' must correspond to a class or package
- Local variable name 'YAxel' must match package name

- Line is longer than 100 characters (found 159). (102:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 122). (130:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 104). (179:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 116). (180:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 107). (181:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 122). (182:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 161). (183:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 138). (184:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 138). (185:0) [LineLength] <Google Checks>
- Line is longer than 100 characters (found 115). (195:0) [LineLength] <Google Checks>

The slide features a dark blue background with decorative hexagonal patterns in the corners. The top-left corner has a single orange hexagon. The top-right corner has a cluster of orange and blue hexagons. The bottom-left corner has a cluster of orange and blue hexagons. The bottom-right corner has a small cluster of orange and blue hexagons.

Static Code Analysis & Clean-Up



Final Style Check Result

The background is a dark navy blue. In the four corners, there are decorative patterns of hexagons. The top-left corner has a single orange hexagon. The top-right corner has a cluster of hexagons, some orange and some light blue. The bottom-left corner has a cluster of hexagons, some orange and some light blue. The bottom-right corner has two hexagons, one orange and one light blue.

Security & Code Quality Improvements

✓ StockMaster3000Saed (found 44 bug items in 50 classes)

> Malicious code vulnerability (37 items)

✓ Dodgy code (1 item)

✓ Dubious method invocation (1 item)

✓ Code contains a hard coded reference to an absolute path

© Hard coded reference to an absolute path

> Performance (6 items)

> Malicious code vulnerability (37 items)

> Dodgy code (1 item)

✓ Performance (3 items)

✓ Unused field (1 item)

✓ Unused field (1 item)

© Unused field: org.stockmaster3000.stockmaster3000

✓ Unread field (2 items)

✓ Unread field (2 items)

© Unread field: org.stockmaster3000.stockmaster3000

© Unread field: org.stockmaster3000.stockmaster3000

Unread field: org.stockmaster3000.stockmaster3000

main > java > org > stockmaster3000 > stockmaster3000

Preview DashboardTab.java:

```
105 CategoryService categoryService;
106 this.headerComponent = headerComponent;
107 this.securityService = securityService;
108 this.inventoryService = inventoryService;
109 this.productService = productService;
110 this.categoryService = categoryService;
111 this.supplierService = supplierService;
```

StockMaster3000Saed (found 37 bug items in 40 classes)

✓ Malicious code vulnerability (34 items)

> Method returning array may expose internal representation

✓ Storing reference to mutable object (22 items)

✓ May expose internal representation by incorporating reference

© org.stockmaster3000.stockmaster3000.model.CategoryService

© org.stockmaster3000.stockmaster3000.model.InventoryService

© new org.stockmaster3000.stockmaster3000.model.InventoryService

© org.stockmaster3000.stockmaster3000.model.InventoryService

© org.stockmaster3000.stockmaster3000.model.InventoryService

© org.stockmaster3000.stockmaster3000.model.ProductService

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© org.stockmaster3000.stockmaster3000.model.ReportService

© org.stockmaster3000.stockmaster3000.model.SupplierService

© org.stockmaster3000.stockmaster3000.model.UserService

© org.stockmaster3000.stockmaster3000.tab.DashboardTab

© new org.stockmaster3000.stockmaster3000.tab.DashboardTab

© new org.stockmaster3000.stockmaster3000.tab.InventoryTab

© new org.stockmaster3000.stockmaster3000.tab.ProductTab

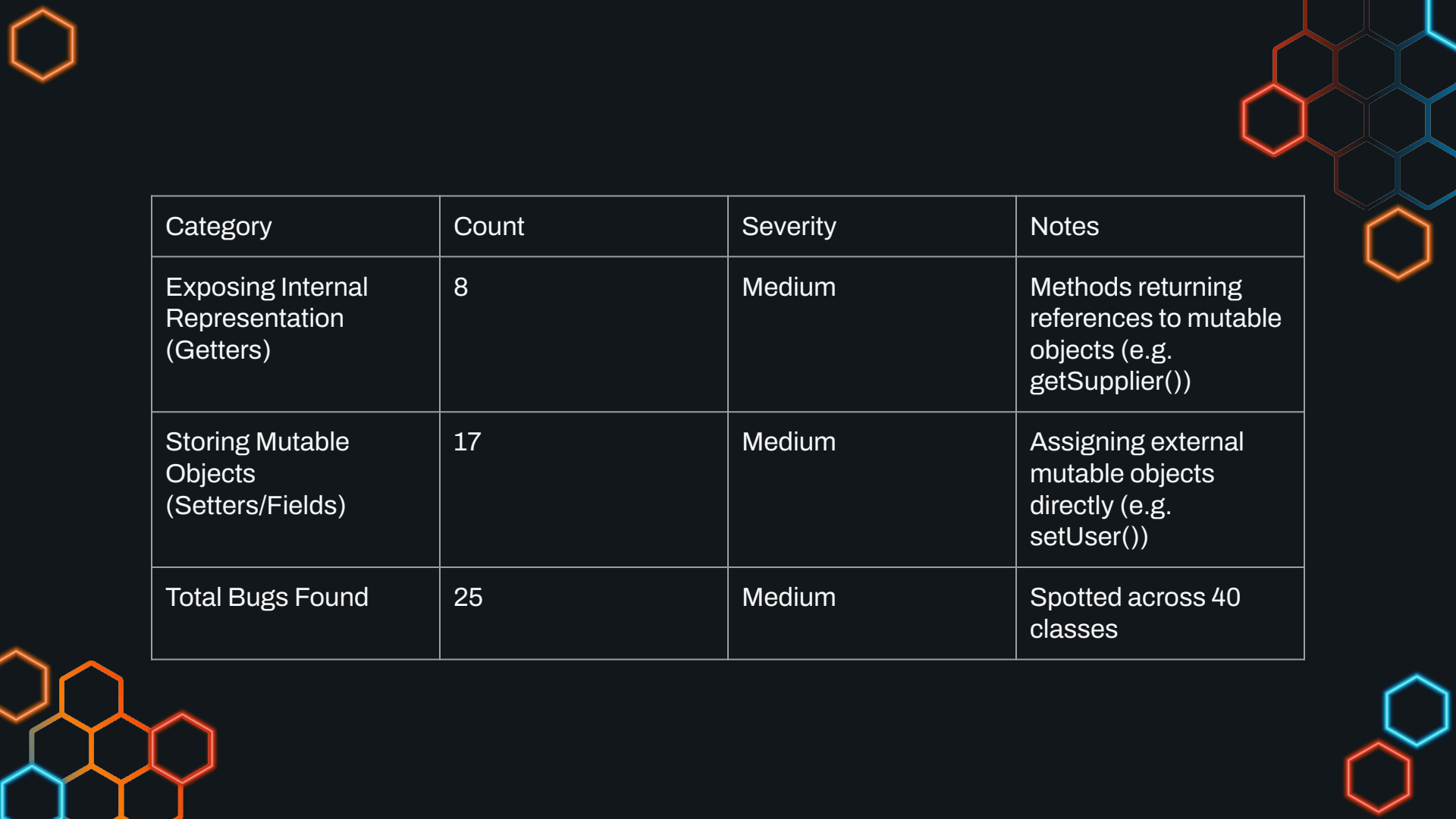
© new org.stockmaster3000.stockmaster3000.tab.ReportTab

© new org.stockmaster3000.stockmaster3000.tab.SupplierTab

© new org.stockmaster3000.stockmaster3000.tab.UserService

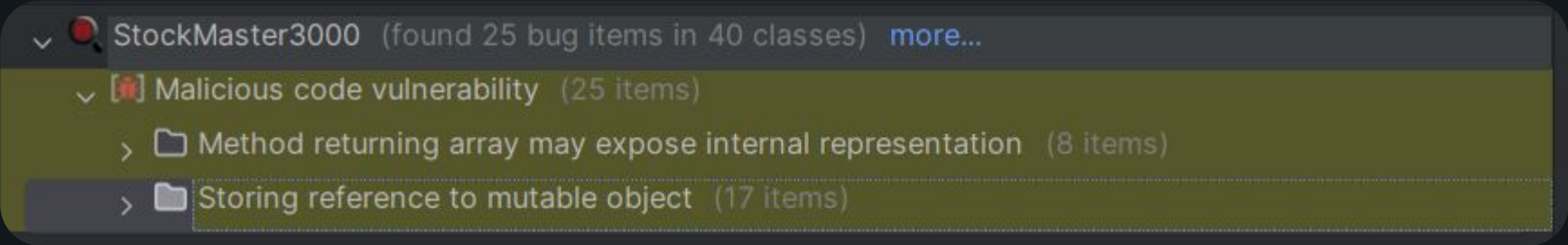
Preview OpenAiClient.java:

```
55 String absPath = "C:/Users/Käyttaja/Desktop/StockMaster3000/StockMaster3000";
56  if (Files.exists(Paths.get(absPath))) {
57     dotenv = Dotenv.configure()
58         .directory(path: "C:/Users/Käyttaja/Desktop/StockMaster3000/StockMaster3000")
59         .ignoreIfMissing()
60         .load();
61
62     return dotenv;
63 }
64
```

Category	Count	Severity	Notes
Exposing Internal Representation (Getters)	8	Medium	Methods returning references to mutable objects (e.g. <code>getSupplier()</code>)
Storing Mutable Objects (Setters/Fields)	17	Medium	Assigning external mutable objects directly (e.g. <code>setUser()</code>)
Total Bugs Found	25	Medium	Spotted across 40 classes

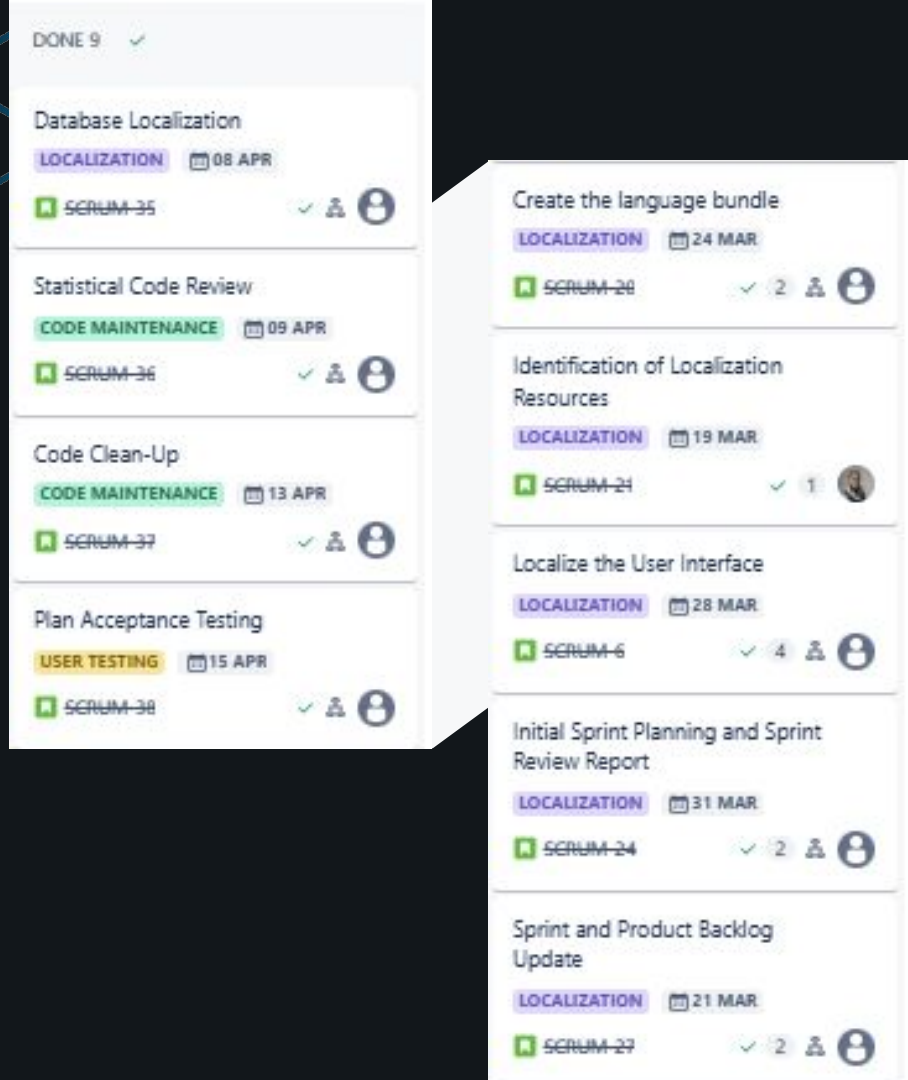
Final SpotBugs Result

- 
- StockMaster3000 (found 25 bug items in 40 classes) [more...](#)
- Malicious code vulnerability (25 items)
 - Method returning array may expose internal representation (8 items)
 - Storing reference to mutable object (17 items)
- The image shows a screenshot of a SpotBugs analysis result for a project named 'StockMaster3000'. The results are displayed in a dark-themed window. The top line shows the project name, a magnifying glass icon, and the text '(found 25 bug items in 40 classes)' followed by a 'more...' link. Below this, a tree view shows a category 'Malicious code vulnerability' with a red bug icon and '(25 items)'. Under this category, two sub-items are listed: 'Method returning array may expose internal representation' with '(8 items)' and 'Storing reference to mutable object' with '(17 items)'. The 'Storing reference to mutable object' item is currently selected, highlighted with a light blue background.

Jira

Backlog Status

- User Stories: 9
- User Story Points: 22



The screenshot displays a Jira backlog with several tasks, each with a status, a category, a due date, and a progress indicator. The tasks are:

- DONE 9** (Status: DONE, Progress: 9/9) - Checkmark icon.
- Database Localization** (Category: LOCALIZATION, Due Date: 08 APR, Progress: 35/35) - Checkmark icon.
- Statistical Code Review** (Category: CODE MAINTENANCE, Due Date: 09 APR, Progress: 36/36) - Checkmark icon.
- Code Clean-Up** (Category: CODE MAINTENANCE, Due Date: 13 APR, Progress: 37/37) - Checkmark icon.
- Plan Acceptance Testing** (Category: USER TESTING, Due Date: 15 APR, Progress: 38/38) - Checkmark icon.
- Create the language bundle** (Category: LOCALIZATION, Due Date: 24 MAR, Progress: 28/28) - Checkmark icon.
- Identification of Localization Resources** (Category: LOCALIZATION, Due Date: 19 MAR, Progress: 24/24) - Checkmark icon.
- Localize the User Interface** (Category: LOCALIZATION, Due Date: 28 MAR, Progress: 6/6) - Checkmark icon.
- Initial Sprint Planning and Sprint Review Report** (Category: LOCALIZATION, Due Date: 31 MAR, Progress: 24/24) - Checkmark icon.
- Sprint and Product Backlog Update** (Category: LOCALIZATION, Due Date: 21 MAR, Progress: 27/27) - Checkmark icon.

Hours Contributed by Team

Tuesday	2	2	-	1
Wednesday	-	-	2	-
Thursday	-	-	-	1
Friday	3	-	-	-
Saturday	2	-	-	-
Sunday	-	-	-	-
Monday	-	-	-	-
Tuesday	1	-	-	-
Wednesday	-	1	-	-
Thursday	2	-	-	2
Friday	2	3	-	-
Saturday	2	-	-	-
Sunday	-	1	-	-
Monday	-	6	8	-

Sprint 6 hours

Viet

Pavel

Ivan

Saed

14

13

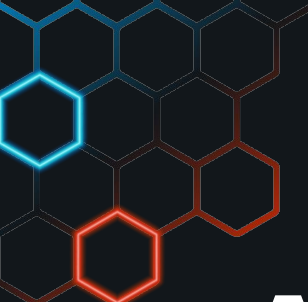
12

10



Challenges Faced

- **SpotBugs** mutable object warnings were tricky and time-consuming to resolve
- 
- 
- 



Ai Tools Used

- **ChatGPT / DeepSeek** = Assisted in refactoring code, writing Javadocs, and resolving static analysis issues
 - **NapkinAI** = Visuals for Presentations
- 
- 



Thanks!

